

An Empirical Study of Data Free Weight Quantization on MedMNIST

Abhishek Shekhar Abhey Singh Guram

Abstract

Weight-only post-training quantization can reduce a model’s stored representation without calibration inputs, but low reconstruction error need not preserve classification performance. We study MobileNetV3-Small on four MedMNIST datasets with three training seeds each. A matched four-bit experiment crosses random orthogonal rotation with uniform and Gaussian scalar families. We then save and decode actual packed representations using FP16 scales and excluded parameters, export per-example predictions, and evaluate a restricted weight-only DFQ adaptation. The saved INT8 representation changes dataset-level mean AUC by at most 0.0018 relative to FP32. Saving at the specified precision changes individual four-bit AUCs by at most 0.00282 relative to the original full-scale reconstruction. Among the saved four-bit configurations, the DFQ adaptation has the largest observed mean AUC on BloodMNIST and PathMNIST, while rotated uniform has the largest means on DermaMNIST and PneumoniaMNIST. Paired image-bootstrap intervals quantify test-set uncertainty conditional on the three checkpoints; training-seed variability remains substantial. The tested two- and three-bit configurations degrade sharply. Rotation sensitivity and retrospective weight-error selection show that favorable weight reconstruction does not reliably predict downstream performance. These results establish storage fidelity for the implemented decoder and runtime, without demonstrating inference acceleration or clinical utility.

1 Introduction

Post-training quantization reduces parameter precision after training [Gholami et al., 2022]. Many effective methods use representative, usually unlabeled, inputs to estimate activation statistics or optimize reconstruction [Nagel et al., 2020; Li et al., 2021]. Such calibration can be inexpensive when inputs are available locally. The narrower question considered here is what a model recipient can establish when representative inputs are unavailable and compression must use stored model information alone.

This is an empirical measurement study of a compact convolutional network, rather than a new quantization algorithm. MobileNetV3-Small is designed as a

lightweight architecture [Howard et al., 2019]; the checkpoints studied here are floating-point fine-tuned models, not previously low-bit quantized models. MedMNIST provides standardized biomedical image-classification tasks and fixed splits [Yang et al., 2023]. Their role is to support controlled comparisons across four datasets, not to establish clinical performance. We reconstruct rounded weights in FP32 and retain floating-point activations, isolating weight distortion from integer-kernel effects.

The study tests three questions. First, how much classification performance is retained by simple weight rounding at different bit widths? Second, at matched cardinality, how do random rotation and scalar-quantization family affect four-bit behavior? Third, how sensitive are results to training and rotation seeds, and can a weight-only reconstruction criterion identify a useful rotation? The contribution is a checkpoint-paired characterization of these effects, together with an audit of the assumptions behind storage estimates. Fixed experimental settings and traceable artifacts support this measurement goal [Pineau et al., 2021].

A smaller stored representation could be useful for model distribution or archiving, but a roughly six-megabyte backbone alone does not establish a consequential deployment saving. The practical implication of the present evidence is therefore limited: favorable numerical compression ratios and reconstruction error should be checked against downstream metrics, variability, and an actual serialization round trip before being used as deployment evidence.

2 Related work

Calibration-based PTQ includes adaptive rounding and block reconstruction [Nagel et al., 2020; Li et al., 2021]. The journal treatment by Gong et al. [2025] combines BRECQ and QDrop in a framework addressing very low precision. GPTQ, AWQ, and SmoothQuant likewise use information derived from calibration inputs in their respective language-model settings [Frantar et al., 2023; Lin et al., 2024; Xiao et al., 2023]. Greedy path-following quantization is another retraining-free approach [Lybrand and Saab, 2021]; retraining-free does not itself mean input-free. Quantization-aware training demonstrates that extremely low precision can succeed under a different optimization protocol

[Hubara et al., 2018]. Consequently, failure of the simple two- and three-bit configurations tested here is not a general impossibility result for low-bit networks.

Data-free quantization covers several distinct information-access regimes. DFQ uses weight equalization and bias correction [Nagel et al., 2019]. ZeroQ and Diverse Sample Generation synthesize surrogate calibration inputs from model information [Cai et al., 2020; Qin et al., 2023]. In healthcare segmentation, DFQ-SAM synthesizes inputs and calibrates a Segment Anything Model without original images [Li et al., 2026]. DFQ-SAM differs from this study in architecture, task, and synthetic calibration. Our follow-up evaluates a restricted weight-only DFQ adaptation using stored BN statistics, while excluding both real and synthetic calibration inputs. This is not a reproduction of canonical W8A8 DFQ and cannot establish superiority over data-free PTQ generally.

Transform quantization for CNN compression jointly considers transforms, rate, and distortion [Young et al., 2022]. QuIP applies incoherence processing alongside an error-aware quantizer for language models [Chee et al., 2023], while OptRot learns data-free rotations for that architecture class [Gadhikar et al., 2025]. TurboQuant motivates the random-rotation and Gaussian-codebook reconstruction mechanism used here, originally in online vector quantization [Zandieh et al., 2025]. We adapt that mechanism to convolutional weight rows without reproducing the complete algorithms or deployment settings of these methods. NUPES explores a different nonuniform quantizer based on optimized power transforms [Yvinec et al., 2025]. These precedents make scalar-family and transform comparisons relevant; they do not predict which combination will preserve a particular classifier.

Empirical comparisons also require uncertainty to be separated from observed rankings. Benavoli et al. [2017] discuss limitations of conventional classifier-comparison testing, and Pineau et al. [2021] emphasize reproducible evidence. We use paired differences on fixed checkpoints, report training-seed variability, and treat the small sample as a limit on inference. These citations motivate reporting discipline; no Bayesian comparison procedure is claimed to have been performed.

3 Methods

3.1 Datasets and checkpoints

We evaluate DermaMNIST, PneumoniaMNIST, BloodMNIST, and PathMNIST using the 64×64 MedMNIST variants and their supplied train,

validation, and test splits (Table 1). Inputs are divided by 255 and normalized using ImageNet channel means and standard deviations. For grayscale inputs, both normalization vectors are averaged across channels. The first convolution is adapted to one input channel by averaging its pretrained RGB filters; the classifier head is separately replaced to match the task. The model uses ImageNet-pretrained MobileNetV3-Small weights before fine-tuning.

The archived training implementation uses cross-entropy loss, AdamW with weight decay 10^{-4} , cosine learning-rate decay, batch size 128, and best-validation-AUC checkpoint selection, with later epochs winning ties. The original manuscript reports a 12-epoch budget; the script defaults to 15 epochs and learning rate 10^{-3} . The checkpoints do not store these run arguments. Relative to the cached ImageNet V1 checkpoint’s BN counter of 699,033, all saved counters are consistent with selected epochs 6–12 under the batch-128 training loop. Session records of the two training launches confirm the 12-epoch budget and the 64-pixel input size; the learning rate is not recorded in them and is taken to be the script default of 10^{-3} (see the repository’s training_provenance directory). Seeds are 42, 123, and 456; all 12 checkpoint hashes match the corrected manifest. Validation-based selection belongs to training, before the data-free compression stage.

AUC is the primary ranking metric. The implementation computes binary ROC AUC from the positive-class softmax probability and multiclass AUC as an unweighted macro average of one-versus-rest ROC AUCs. Accuracy and macro F1 use the softmax argmax. Class imbalance motivates examining class-sensitive metrics alongside accuracy [Zhang et al., 2023]; AUC alone does not establish minority-class performance, probability calibration, or clinical usefulness.

3.2 Tensor policy and evaluated precision

A convolutional weight tensor with shape (o, c, k_h, k_w) is reshaped into o rows of length $n = c k_h k_w$. Linear weights already have this form. Convolutional and linear weight tensors are eligible when they contain at least 256 parameters. For low-bit evaluation, the first and last eligible weight tensors in module order use INT8; other eligible tensors use the selected bit width. Rotation is disabled when $n < 64$, while the scalar family is retained.

For the three RGB datasets, the protected tensors are the first convolution and final classifier. For PneumoniaMNIST, the 144-parameter grayscale first

convolution is excluded; the first protected tensor is features.1.block.2.0.weight, followed by the final classifier. This distinction follows the implementation and is applied without consulting compression-time validation performance.

The original full-scale reconstruction differs from the saved representation. Its quantizers reconstruct to FP32 without first rounding scales to FP16, and excluded parameters remain at checkpoint precision. Sections 4.1–4.5 report that original experiment; Section 4.6 reports a separate completed serialization round trip with FP16 scales and excluded parameters. This distinction keeps the earlier rotation-sensitivity analyses traceable while directly testing the stated storage precision.

3.3 Scalar families and rotations

Both low-bit scalar families have $L = 2^b$ reconstruction levels. For a weight row w , the uniform family uses $a = \max(\max_i |w_i|, 10^{-12})$, scale $s_U = a/(L - 1)$, and integer codes $C_\beta = \{-(L - 1), -(L - 3), \dots, L - 3, L - 1\}$. Each coordinate is reconstructed at its nearest scaled level. The implementation sends zero to the smallest positive level and nonzero midpoint ties away from zero. Thus the grid is symmetric and has no exact-zero level.

The Gaussian family uses $s_G = \max(\|w\|_2, 10^{-12})/\sqrt{n}$ and quantizes w/s_G to a shared L -level codebook. The codebook is fitted by the Lloyd–Max procedure [Lloyd, 1982; Max, 1960], using two million standard-normal samples, seed 0, and at most 100 iterations. It is a finite-sample approximation to a Gaussian-optimal scalar codebook, not a dataset-specific calibration. Nearest-centroid assignments use midpoint boundaries with lower-index tie breaking.

A Haar-random orthogonal matrix Π is generated per eligible tensor by QR decomposition of an independent Gaussian matrix with diagonal-sign correction. Rotated uniform quantization applies the uniform rule to $w\Pi^T$ and then multiplies the reconstructed row by Π . Rotated Gaussian quantization forms $y = (w/s_G)\Pi^T$, assigns each coordinate to a Gaussian centroid, and reconstructs $\hat{w} = s_G \hat{y}\Pi$. Orthogonality preserves norms before quantization; approximate Gaussian coordinates motivate the Gaussian codebook but do not guarantee downstream classification preservation.

The main rotation seed is 0. Tensor i receives seed $(1,000,003r + 97,409i + 17)$ modulo $(2^{32} - 5)$, where r is the rotation seed and i is the zero-based eligible-tensor index. Training seeds do not enter this mapping. A separate experiment uses $r \in \{0, 1, 2, 3, 4\}$ for the four-bit rotated Gaussian method. The four-bit ablation crosses rotation off/on with

uniform/Gaussian family. Cardinality and tensor policy are matched, but the two families also differ in scale definition; the comparison is therefore between scalar families, not an isolated test of centroid spacing.

3.4 Paired analysis and retrospective rotation selection

Dataset-level metrics are means and sample standard deviations over three training seeds. Each method contrast uses checkpoint-paired AUC differences. We additionally report exploratory 95% Student-t intervals, $\bar{\Delta} \pm 4.303 s\Delta/\sqrt{3}$, with two degrees of freedom. These intervals assume independent, approximately normal training-seed differences conditional on a fixed dataset and test set; that assumption cannot be assessed reliably with three seeds. They are not test-example bootstrap intervals or simultaneous intervals across comparisons.

As a sensitivity check, two-sided sign-flip calculations enumerate all eight sign assignments of each three-checkpoint difference vector. Under a symmetric null distribution, the smallest attainable two-sided p-value is 0.25. No reported contrast reaches 0.05 by this check. The analyses are exploratory, with no multiple-comparison claim or statistically established ordering. Separate marginal error-bar overlap is not used as a significance test, and checkpoints from different datasets are not treated as identically distributed replications.

To examine the reconstruction-based selection suggestion, we retrospectively reapply the recorded four-bit Gaussian quantizer to all five rotation seeds for each checkpoint. The selector minimizes the sum of squared reconstruction errors across all eligible weight tensors, divided by their total squared weight norm. One global rotation seed is selected per checkpoint, with the smallest seed breaking ties. Selection uses only weights; the selected seed is then joined to its already recorded downstream evaluation. Because the rule was added after inspecting the original study, this is a retrospective analysis rather than a prospectively held-out confirmation.

3.5 Storage accounting and decode work

The original size function sums nominal parameter bits and divides by 8×10^6 . It omits model buffers, container overhead, codebooks, and rotation metadata, and even the reported FP32 baseline charges 16 bits for excluded parameters. Consequently, the original values near 6.06, 1.56, and 0.810 MB are neither complete FP32 state sizes nor measured compressed files.

We first audit a specified payload analytically from checkpoint tensor shapes. Each eligible tensor t contributes $\lceil b_t N_t / 8 \rceil$ bytes of indices and $2o_t$ bytes of FP16 scales; excluded parameters contribute 2 bytes each. Buffers retain their original precision. Byte rounding is applied independently per tensor. Codebooks, seeds and container headers are additional. Table 6 reports this analytical payload; Table 9 reports measured complete file sizes from the separate saved-model experiment.

For o rows of length n , inverse dense rotation costs $o n(2n - 1)$ scalar multiply/add operations, excluding scaling and codebook lookup. Regenerating each matrix by QR adds $O(n^3)$ setup work and $O(n^2)$ temporary storage. A padded fast Hadamard transform at dimension d equal to the smallest power of two at least n instead uses $o d \log_2 d$ additions/subtractions, before normalization and sign operations. Padding changes the representation and potentially the rate; the counts compare transform work, not measured speed or equal-accuracy codecs.

3.6 Saved representation and matched DFQ verification

Sections 4.1–4.5 retain the original full-scale reconstruction experiments for traceability. The follow-up in Section 4.6 evaluates the actual stored representation on the same 12 checkpoints. Coordinate indices are bit-packed at the assigned width; each output row stores an IEEE FP16 scale. Excluded parameters, including biases and BN affine parameters, are stored in FP16, while buffers retain their original dtypes. Gaussian centroids and per-tensor rotation seeds are included in file metadata. The decoder reads only saved bytes and regenerates rotations using the recorded NumPy QR implementation. Every decoded state is loaded into a fresh model before evaluation. Reported precision effects include both scale rounding and excluded-parameter rounding; they do not isolate scale rounding alone.

The DFQ-style baseline folds Conv–BN pairs, equalizes 13 safe Conv–ReLU–Conv paths, and applies analytical bias correction to the two pointwise convolutions immediately following BN/ReLU in residual blocks 2 and 3. Nine equalization paths are the SE $fc1/fc2$ pairs; four are within the two ReLU residual blocks. No scaling crosses hard-swish, residual additions, or SE multiplication. Bias correction uses rectified-normal means inferred from stored BN parameters and corrects local decoded-weight error. The corrected biases are then stored in FP16 without requantizing weight codes. High-bias absorption and activation

quantization are omitted. Thus the baseline is an explicit weight-only adaptation of Nagel et al. [2019], not canonical W8A8 DFQ. A BN-folding-only control uses the same quantizer, layer eligibility, protected layers, and storage precision. Folding changes parameter and buffer counts, so this is a matched quantization-policy comparison rather than an exactly equal byte-budget comparison.

The follow-up uses Python 3.12.14, PyTorch 2.14.0, NumPy 2.5.3, and MPS inference with batch size 256. Only test-split arrays are loaded for scoring; compression receives no input examples or labels. Per-example logits, probabilities, labels, split indices, and checkpoint/test-data hashes are exported. A class-stratified paired bootstrap uses 2,000 replicates and seed 20260910, sharing image indices across methods and training seeds. Metrics are recomputed per checkpoint and then averaged for the fixed-three-checkpoint estimand. Percentile 95% intervals are marginal, conditional on the trained models and observed class counts, and not adjusted for multiple comparisons. They quantify image-sampling uncertainty separately from the three-seed t intervals.

4 Results

4.1 INT8 rounding retains the observed baseline metrics

The absolute dataset-level mean AUC change is at most 0.002, and the largest absolute checkpoint-level change is 0.006 on PneumoniaMNIST. These are observed differences, not an equivalence test. FP16 rounding of eligible weights also closely reproduces FP32 metrics. The simulations support limited sensitivity to higher-bit weight rounding on these checkpoints, but do not validate the precision assumptions of the nominal storage representation.

4.2 Synthetic reconstruction and four bit classification

Figure 1 compares normalized squared reconstruction error on 256×512 synthetic Gaussian and Student- t matrices, with three degrees of freedom for the latter. The synthetic generator uses seed 2026 and the rotations use seed 17. Rotation has little visible effect for Gaussian weights and lowers error for heavy-tailed weights. The rotated Gaussian family has lower reconstruction error than rotated uniform in the displayed cases. Unrotated Gaussian quantization is not uniformly better than unrotated uniform; the heavy-tailed four-bit case illustrates this distinction.

Rotated uniform has the largest observed mean AUC on all four datasets (Table 3). Relative to unrotated

uniform, the mean paired changes are +0.059, +0.155, +0.041, and +0.069 for DermaMNIST, PneumoniaMNIST, BloodMNIST, and PathMNIST, respectively. Only eight of the twelve individual checkpoints improve. Rotation improves Gaussian-family AUC on all twelve checkpoints, but neither that consistency nor the dataset-level ordering establishes a general method advantage.

All rotated-uniform versus rotated-Gaussian intervals include zero. The DermaMNIST Gaussian rotation effect has a positive t interval, but its exact sign-flip p-value is 0.25, demonstrating the sensitivity of inference to assumptions at $n = 3$. We retain the observed effect estimates without a significance or superiority claim.

AUC also conceals substantial losses in class prediction. For rotated uniform, mean accuracy is 0.574, 0.757, 0.496, and 0.532 across the four datasets in Table 1 order; corresponding macro F1 values are 0.203, 0.682, 0.376, and 0.466. FP32 mean accuracy ranges from 0.765 to 0.971. Thus partial retention of ranking ability does not imply acceptable classification decisions.

4.3 Two and three bit configurations

Averaged equally over twelve checkpoints, uniform and rotated-Gaussian AUCs are 0.498 and 0.466 at two bits, 0.505 and 0.521 at three bits, and 0.723 and 0.734 at four bits. The INT8 average is 0.966. These are averages of checkpoint-level AUCs, not AUCs computed from pooled predictions across tasks. The two- and three-bit conclusions concern only the two evaluated families and fixed layer policy; rotated uniform was not evaluated at those bit widths.

4.4 Rotation sensitivity and weight only selection

PneumoniaMNIST has the largest mean within-checkpoint AUC SD, 0.075. Its checkpoint trained with seed 42 ranges from 0.607539 to 0.929602 across rotations. Training-related variability also remains substantial after averaging over rotations (Table 5). The factorial sweep concerns the Gaussian method; it does not establish the same sensitivity for rotated uniform.

Minimum-weight-error selection improves AUC on five checkpoints, worsens it on six, and leaves one unchanged. Mean AUC changes are +0.039 on DermaMNIST, +0.048 on PneumoniaMNIST, -0.031 on BloodMNIST, and -0.002 on PathMNIST. Selected seeds are 1 for all DermaMNIST, BloodMNIST, and PathMNIST checkpoints; for PneumoniaMNIST seeds 42, 123, and 456, they are 0, 3, and 3. This weight-only

rule is implementable without calibration, but it does not reliably improve classification. Test AUC must not be used to choose a replacement seed in a data-free deployment.

4.5 Audited payload and transform costs

The analytical payload accounting gives approximately 6.13–6.16 MB for original state, 1.60–1.61 MB for INT8, and 0.854–0.862 MB for four bits before metadata. Section 4.6 separately reports actual saved-file sizes and decoded-model metrics. Dense inverse rotations require 1,142,731,008 scalar multiply/add operations per model under the implemented policy, excluding QR regeneration, lookup, and scaling. A padded Hadamard alternative would require 21,303,296 transform additions/subtractions before normalization and sign operations. This arithmetic comparison demonstrates neither a latency ratio nor preserved accuracy.

4.6 Saved model verification and matched comparator

All 132 archived-precision reruns reproduce the published six-decimal metrics to rounding tolerance (maximum absolute discrepancy 5×10^{-7}). The study evaluates 228 saved model files and exports 384 prediction files including controls. Recomputing metrics from the exports agrees within 1.2×10^{-16} , and all packed/prediction hashes verify. Across the 132 saved-versus-archive pairs, maximum absolute changes are 0.00381 AUC, 0.00962 accuracy, and 0.00844 macro F1; restricted to four bits, they are 0.00282, 0.00649, and 0.00403. FP16 scale underflow is retained literally, with at most 86 row scales rounding to zero in a file; all decoded tensors and predictions remain finite. The empirical round trip therefore matters even when aggregate conclusions remain similar.

DFQ* has the largest observed four-bit mean AUC on BloodMNIST (0.831) and PathMNIST (0.916); rotated uniform leads on DermaMNIST (0.673) and PneumoniaMNIST (0.840). DFQ* improves on the saved unrotated uniform baseline by +0.010, +0.048, +0.061, and +0.096 AUC in Table 1 dataset order. Folding-only comparisons yield similar improvements, so folding alone does not explain this pattern. The adaptation’s limited scope and large training-seed SDs preclude a general method ranking.

For these fixed checkpoints, the DFQ*-uniform image intervals exclude zero on PneumoniaMNIST, BloodMNIST, and PathMNIST but include zero on DermaMNIST. This is compatible with wide training-seed intervals: the estimands differ. Saved INT8 minus FP32 mean AUC changes are -0.00038, +0.00175,

-0.00008, and -0.00060 in Table 1 dataset order. Several tiny differences have image intervals excluding zero; statistical detectability should not be confused with a prespecified practical-equivalence claim.

5 Discussion and limitations

The evidence separates reconstruction fidelity, downstream classification, and deployable compression. A distribution-matched codebook can perform well on synthetic weight MSE while a uniform family has larger observed downstream AUC. The retrospective selection result closes part of this gap: choosing among actual checkpoint rotations by weight error still produces mixed AUC outcomes. It does not establish that all reconstruction-based selectors fail, because only one aggregate criterion and five candidates were tested.

A fixed random seed is compatible with data-free execution; unpredictable downstream performance is the limitation. The experiments characterize this risk rather than establish a reliable low-bit deployment method. INT8 is substantially less disruptive, including after actual storage and decoding, but no prespecified equivalence margin was tested. Small observed changes should therefore not be described as formal equivalence or universal accuracy preservation.

The restriction to no real or synthetic calibration inputs is not a general privacy guarantee or a claim that local calibration is infeasible. The matched DFQ adaptation improves the empirical context and demonstrates that a relevant data-free comparator can change the observed ranking. Its limited activation compatibility and two-layer analytical bias correction leave a gap to the full DFQ procedure. Calibration-based methods would quantify the cost of the input restriction, but belong to a different information-access regime.

The study is limited to one architecture, four standardized datasets, one image size, three training seeds, and five Gaussian rotation seeds. There are no external-site evaluations, patient-level clinical analyses, or operating-threshold assessments. The paired image bootstrap addresses finite-test-set uncertainty conditional on these checkpoints and the observed class counts. It neither replaces additional training seeds nor captures patient or acquisition clustering, whose identifiers are unavailable in the exported benchmark arrays.

Weights are reconstructed to FP32 and activations remain floating point. No integer kernels, inference-memory, latency, or energy measurements were performed. The saved-file experiment verifies the implemented representation in the recorded

environment, including FP16 underflow; it does not establish portable rotation regeneration across numerical libraries. The learning rate used in training is inferred from the script default rather than a primary record. New code and prediction artifacts must accompany publication for these follow-up results to be reproducible.

6 Conclusion

In this checkpoint-paired MedMNIST study, actual saved INT8 representations change mean AUC modestly, while the tested two- and three-bit settings substantially degrade ranking performance. Four-bit results depend on dataset and method: the restricted DFQ adaptation and rotated uniform have the largest observed means on different datasets. Paired test-image intervals and training-seed variability describe different uncertainty sources and do not establish a universal winner. The completed storage round trip, documented prediction format, and explicit baseline adaptations make the numerical findings auditable while leaving deployment and clinical questions open.

Data and code availability

The original experiment code, checkpoints, aggregate results, and manifests are available at <https://github.com/Abhi183/no-peeking-dfq-medmnist>. The original analyses use `dfq_corrected_fixed.csv` and `dfq_rotation_factorial.csv`, rather than validation-selected results. The accompanying revision artifacts add the packed codec, matched DFQ adaptation, evaluation runner, 384 indexed prediction files, paired-bootstrap results, and dependency/hash manifests. These revision artifacts are in the repository; the packed model files, prediction files, and bootstrap replicates are attached to its storage-study-v1 release. MedMNIST supplies the benchmark data [Yang et al., 2023].

Tables and Figures

Table 1. Dataset characteristics. Imbalance is the largest-to-smallest training-class ratio.

Dataset	Classes	Train	Validation	Test	Imbalance
DermaMNIST	7	7,007	1,003	2,005	58.7:1
PneumoniaMNIST	2	4,708	524	624	2.9:1
BloodMNIST	8	11,959	1,712	3,421	2.7:1
PathMNIST	9	89,996	10,004	7,180	1.6:1

Table 2. FP32 and INT8 test metrics, mean $\pm$ sample SD across three training seeds. INT8 refers to weight-rounding simulation.

Dataset	FP32 AUC	INT8 AUC	FP32 accuracy	INT8 accuracy
DermaMNIST	0.9170 $\pm$ 0.0046	0.9166 $\pm$ 0.0058	0.7651 $\pm$ 0.0197	0.7619 $\pm$ 0.0172
PneumoniaMNIST	0.9575 $\pm$ 0.0081	0.9590 $\pm$ 0.0054	0.8467 $\pm$ 0.0298	0.8446 $\pm$ 0.0167
BloodMNIST	0.9985 $\pm$ 0.0002	0.9984 $\pm$ 0.0002	0.9708 $\pm$ 0.0025	0.9698 $\pm$ 0.0021
PathMNIST	0.9906 $\pm$ 0.0009	0.9900 $\pm$ 0.0013	0.8983 $\pm$ 0.0033	0.8953 $\pm$ 0.0016

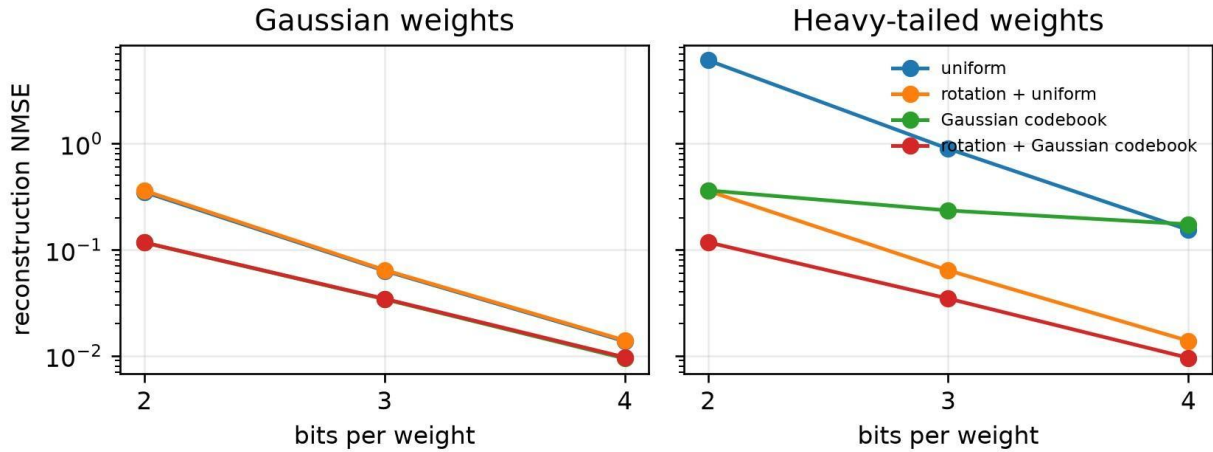


Figure 1. Synthetic normalized reconstruction MSE at two, three, and four bits. Uniform and Gaussian families are each shown with and without rotation. These controlled weight distributions are not learned checkpoint weights.

Table 3. Matched four-bit ablation, test AUC mean $\pm$ sample SD over three training seeds. Observed means do not establish a population ranking.

Dataset	Uniform	Rotated uniform	Gaussian	Rotated Gaussian
DermaMNIST	0.614 $\pm$ 0.024	0.673 $\pm$ 0.040	0.546 $\pm$ 0.068	0.595 $\pm$ 0.070
PneumoniaMNIST	0.685 $\pm$ 0.085	0.840 $\pm$ 0.054	0.689 $\pm$ 0.129	0.791 $\pm$ 0.139
BloodMNIST	0.771 $\pm$ 0.018	0.812 $\pm$ 0.092	0.631 $\pm$ 0.097	0.796 $\pm$ 0.029
PathMNIST	0.821 $\pm$ 0.098	0.890 $\pm$ 0.053	0.741 $\pm$ 0.055	0.754 $\pm$ 0.065

Table 4. Mean paired AUC effects and exploratory 95% t intervals across three training seeds. RU = rotated uniform; U = uniform; RG = rotated Gaussian; G = Gaussian. Intervals are conditional on each fixed test set and are not multiplicity-adjusted.

Dataset	RU – U	RG – G	RU – RG
DermaMNIST	+0.059 [-0.077, +0.195]	+0.049 [+0.017, +0.082]	+0.078 [-0.110, +0.265]
PneumoniaMNIST	+0.155 [-0.187, +0.496]	+0.102 [-0.136, +0.339]	+0.049 [-0.359, +0.458]
BloodMNIST	+0.041 [-0.190, +0.272]	+0.165 [-0.007, +0.336]	+0.016 [-0.283, +0.314]
PathMNIST	+0.069 [-0.089, +0.227]	+0.012 [-0.022, +0.047]	+0.137 [-0.101, +0.374]

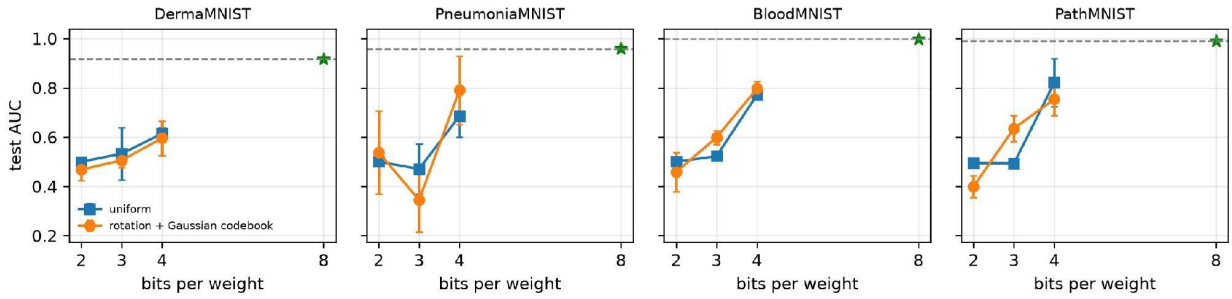


Figure 2. Test AUC for the two prespecified bit-sweep families. Curves show training-seed mean $\pm$ sample SD; dashed lines show FP32 and stars show the separate INT8 baseline. INT8 is not an eight-bit instance of the Gaussian family.

Table 5. Four-bit Gaussian rotation sensitivity and retrospective minimum-weight-error selection. Within-model SD is averaged over the three checkpoints; between-model SD is calculated from rotation-averaged checkpoint AUC. Selection is among five global seeds.

Dataset	Within-model SD	Between-model SD	Fixed seed AUC	Selected AUC	Mean change
DermaMNIST	0.025	0.078	0.595	0.634	+0.039
PneumoniaMNIST	0.075	0.076	0.791	0.839	+0.048
BloodMNIST	0.029	0.041	0.796	0.765	-0.031
PathMNIST	0.022	0.063	0.754	0.752	-0.002

Table 6. Analytical state payloads in decimal MB. FP32 includes parameters and buffers at original precision. INT8 and four-bit estimates assume FP16 scales and excluded parameters, plus original-precision buffers; metadata and container overhead are additional.

Dataset	FP32 payload	INT8 payload	Four-bit payload
DermaMNIST	6.1488	1.6101	0.8596
PneumoniaMNIST	6.1272	1.6047	0.8544
BloodMNIST	6.1529	1.6111	0.8606
PathMNIST	6.1570	1.6121	0.8616

Table 7. AUC after actual storage and decoding, mean $\pm$ sample SD over three training seeds. U and G denote uniform and Gaussian families; R denotes rotation. DFQ* is the restricted weight-only adaptation described in Section 3.6.

Dataset	U	RU	G	RG	DFQ*
DermaMNIST	0.613 $\pm$ 0.025	0.673 $\pm$ 0.039	0.546 $\pm$ 0.069	0.595 $\pm$ 0.070	0.624 $\pm$ 0.075
PneumoniaMNIST	0.684 $\pm$ 0.085	0.840 $\pm$ 0.054	0.688 $\pm$ 0.131	0.790 $\pm$ 0.139	0.732 $\pm$ 0.079
BloodMNIST	0.771 $\pm$ 0.019	0.812 $\pm$ 0.092	0.632 $\pm$ 0.097	0.796 $\pm$ 0.029	0.831 $\pm$ 0.041
PathMNIST	0.821 $\pm$ 0.098	0.890 $\pm$ 0.052	0.741 $\pm$ 0.055	0.754 $\pm$ 0.065	0.916 $\pm$ 0.041

Table 8. Paired AUC effects and 95% image-bootstrap intervals for the fixed-three-checkpoint mean. These conditional intervals do not measure retraining uncertainty or establish equivalence.

Dataset	RG – U	DFQ* – U	DFQ* – folded U
DermaMNIST	-0.018 [-0.044, +0.009]	+0.010 [-0.009, +0.029]	+0.010 [-0.009, +0.029]
PneumoniaMNIST	+0.106 [+0.071, +0.139]	+0.048 [+0.018, +0.076]	+0.047 [+0.018, +0.076]
BloodMNIST	+0.025 [+0.019, +0.032]	+0.061 [+0.055, +0.066]	+0.061 [+0.055, +0.066]
PathMNIST	-0.067 [-0.072, -0.062]	+0.096 [+0.092, +0.099]	+0.095 [+0.092, +0.098]

Table 9. Actual file sizes in decimal MB, averaged over three seeds. Counts include metadata, codebooks, buffers, FP16 scales, and packing padding. These are storage sizes, not inference-memory measurements.

Dataset	FP32	FP16	INT8	RG four bit	DFQ* four bit
DermaMNIST	6.1799	3.1297	1.6454	0.8959	0.8170
PneumoniaMNIST	6.1582	3.1189	1.6400	0.8906	0.8117
BloodMNIST	6.1840	3.1318	1.6465	0.8970	0.8180
PathMNIST	6.1881	3.1338	1.6475	0.8980	0.8190

References

- Benavoli, A., Corani, G., Demšar, J., and Zaffalon, M. (2017). Time for a Change: a Tutorial for Comparing Multiple Classifiers Through Bayesian Analysis. *Journal of Machine Learning Research*, 18(77), 1–36. <https://www.jmlr.org/papers/v18/16-305.html>
- Cai, Y., Yao, Z., Dong, Z., Gholami, A., Mahoney, M. W., and Keutzer, K. (2020). ZeroQ: A Novel Zero Shot Quantization Framework. *CVPR*. <https://arxiv.org/abs/2001.00281>
- Chee, J., Cai, Y., Kuleshov, V., and De Sa, C. (2023). QuIP: 2-Bit Quantization of Large Language Models With Guarantees. *NeurIPS*. <https://arxiv.org/abs/2307.13304>
- Frantar, E., Ashkboos, S., Hoefler, T., and Alistarh, D. (2023). GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. *ICLR*. <https://arxiv.org/abs/2210.17323>
- Gadhikar, A., Grazi, R., and Hensman, J. (2025). OptRot: Mitigating Weight Outliers via Data-Free Rotations for Post-Training Quantization. *arXiv:2512.24124*. <https://arxiv.org/abs/2512.24124>
- Gholami, A., Kim, S., Dong, Z., Yao, Z., Mahoney, M. W., and Keutzer, K. (2022). A Survey of Quantization Methods for Efficient Neural Network Inference. In *Low-Power Computer Vision*, pp. 291–326. Chapman and Hall/CRC. <https://arxiv.org/abs/2103.13630>
- Gong, R., Liu, X., Li, Y., Fan, Y., Wei, X., and Guo, J. (2025). Pushing the Limit of Post-Training Quantization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(7), 5556–5570. <https://doi.org/10.1109/TPAMI.2025.3554523>
- Howard, A., et al. (2019). Searching for MobileNetV3. *ICCV*. <https://arxiv.org/abs/1905.02244>
- Hubara, I., Courbariaux, M., Soudry, D., El-Yaniv, R., and Bengio, Y. (2018). Quantized Neural Networks: Training Neural Networks with Low Precision Weights and Activations. *Journal of Machine Learning Research*, 18(187), 1–30. <https://www.jmlr.org/papers/v18/16-456.html>
- Li, Y., et al. (2021). BRECO: Pushing the Limit of Post-Training Quantization by Block Reconstruction. *ICLR*. <https://arxiv.org/abs/2102.05426>
- Li, Z., Zhang, J., and Gu, Q. (2026). Privacy-Preserving SAM Quantization for Efficient Edge Intelligence in Healthcare. *International Journal of Computer Vision*, 134, article 201. <https://doi.org/10.1007/s11263-026-02791-6>
- Lin, J., et al. (2024). AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. *Proceedings of Machine Learning and Systems*. <https://arxiv.org/abs/2306.00978>
- Lloyd, S. P. (1982). Least Squares Quantization in PCM. *IEEE Transactions on Information Theory*, 28(2), 129–137.
- Lybrand, E., and Saab, R. (2021). A Greedy Algorithm for Quantizing Neural Networks. *Journal of Machine Learning Research*, 22(156), 1–38. <https://www.jmlr.org/papers/v22/20-1233.html>
- Max, J. (1960). Quantizing for Minimum Distortion. *IRE Transactions on Information Theory*, 6(1), 7–12.
- Nagel, M., van Baalen, M., Blankevoort, T., and Welling, M. (2019). Data-Free Quantization Through Weight Equalization and Bias Correction. *ICCV*. <https://arxiv.org/abs/1906.04721>
- Nagel, M., Amjad, R. A., van Baalen, M., Louizos, C., and Blankevoort, T. (2020). Up or Down? Adaptive Rounding for Post-Training Quantization. *ICML*. <https://arxiv.org/abs/2004.10568>
- Pineau, J., et al. (2021). Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program). *Journal of Machine Learning Research*, 22(164), 1–20. <https://www.jmlr.org/papers/v22/20-303.html>
- Qin, H., Ding, Y., Zhang, X., Wang, J., Liu, X., and Lu, J. (2023). Diverse Sample Generation: Pushing the Limit of Generative Data-Free Quantization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10), 11689–11706. <https://doi.org/10.1109/TPAMI.2023.3272925>
- Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., and Han, S. (2023). SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. *ICML*. <https://arxiv.org/abs/2211.10438>
- Yang, J., et al. (2023). MedMNIST v2 – A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. *Scientific Data*, 10, 41. <https://doi.org/10.1038/s41597-022-01721-8>
- Young, S. I., Zhe, W., Taubman, D., and Girod, B. (2022). Transform Quantization for CNN Compression. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(9), 5700–5714. Published online 2021. <https://doi.org/10.1109/TPAMI.2021.3084839>
- Yvinec, E., Dapogny, A., and Bailly, K. (2025). NUPES: Non-Uniform Post-Training Quantization via Power Exponent Search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(11), 10012–10021. <https://doi.org/10.1109/TPAMI.2025.3593987>
- Zandieh, A., Daliri, M., Hadian, M., and Mirrokni, V. (2025). TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate. *arXiv:2504.19874*. <https://arxiv.org/abs/2504.19874>
- Zhang, Y., Kang, B., Hooi, B., Yan, S., and Feng, J. (2023). Deep Long-Tailed Learning: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. <https://arxiv.org/abs/2110.04596>